HE2003 Econometrics I: Introduction

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Why Shall We Study Econometrics?

Economics (Macro/Micro) suggest many important relationships, often with policy implications, but virtually never suggest actual quantitative magnitudes. Examples:

- What is the effect on economic growth of a 1 percentage point increase in interest rates by the central bank? (Macro theory suggests that increase in interest rates will slow down the economy.)
- What is the effect of a lockdown policy on a country's economy?
- What is the effect of reducing class size on student achievement?

For actual policy making, we need to know the precise numbers...

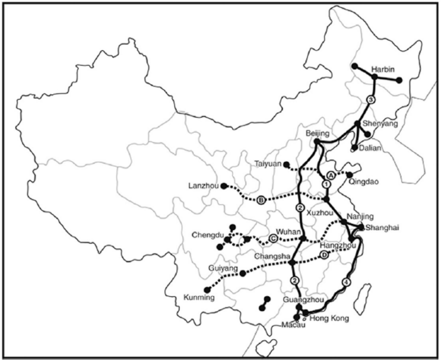
Why Shall We Study Econometrics?

In addition, many economic questions are related to causality or causal effect. Examples:

- "Return to education"? Having large investments in the education industry, the government wants to know: If a person is given another 1 or 2 years of education (e.g., a master's degree), how much will his or her wage increase? (cost/benefit analysis of education)
- Do cigarette taxes reduce smoking? Basic economic theory says that if cigarette prices go up, consumption will go down. But is this true in real world?
- Do high-speed railways promote local economy?

Q: If we observe in data that education level is positively correlated with wage (people with higher education tend to have higher wage), can we conclude that education does have a (positive) causal effect on our earning? (people with higher education level may have higher IQ/ability in the first place)

Why Shall We Study Econometrics?



Map of a high-speed railway system

Q: If we observe that economic performance of cities is positively correlated with having high-speed railway nearby, can we conclude that the high-speed railway has a (positive) causal effect on economic performance?

Why Shall We Study Econometrics?

Correlation vs. Causality

Simply observing a positive or negative relationship between two variables is not enough to establish causality. When X and Y are correlated, there are several possible cases:

- Causality: $X \to Y$;
- Reverse causality: $Y \to X$;
- Existence of certain common factor: $C \to X$ and $C \to Y$.

Let us look at the following example: Do they have causality?

- The sales of beer and the sales of air condtionner have positive correlation!
- The temperature and the sales of beer are correlated!

Econometrics helps us to find true causality (causal effect) in data.

Outline of the Course

Important Materials

- Lecture slides, lecture notes, and tutorial questions

Supplementary Reading

- James H. Stock and Mark W. Watson, Introduction to Econometrics (Third or Fourth Edition)

Method of Instruction

- Lectures (Monday) commence in Week 1.
- Tutorials (Monday) commence in Week 2.

Office Hours

- By Appointment: wang.wj@ntu.edu.sg

Outline of the Course

Prerequisite

- 1st year course in Probability and Statistics
- In particular:
  - Mean (Expectation), Variance, Covariance, Correlation,
  - Distribution of a Random Variable (Normal, Chi-squared),
  - Conditional Mean, Random Sampling,
  - Testing, Confidence Interval, etc.

Example:

The conditional mean of Y given $X = x$ can be written as:

$$E(Y|X = x)$$

Application in Econometrics: Suppose we want to study if there exists gender wage gap in the job market. Let $Y = wage$, $X = gender$, and we are interested in the following formula:

$$E(wage|gender = male) - E(wage|gender = female).$$

Example (continued):

- In statistics and econometrics, it is important to distinguish the difference between population and sample.
- For instance, suppose that you are interested in the distribution of wages in Singapore (e.g., average wage, wage spread among workers). It may be difficult to interview all the workers to obtain the whole distribution.
- Thus, we typically estimate on the basis of a group of randomly selected people (say, 1000, 5000, etc). In this example, the wages of all workers in Singapore is the population, while the wages of the 1000 randomly selected people is the sample.

Outline of the Course

Example (continued):

Here, a key idea is that we view the wage of a worker as a random variable. **This does not mean that his/her wage fluctuates randomly.** What is meant is that the selection procedure is random: He/she might or might not be selected into the sample. Had a different sample be drawn, then the observations would have been different. $\rightarrow$ **Hypothesis testing/Confidence interval**

Population (left) vs Sample (right)

Outline of the Course

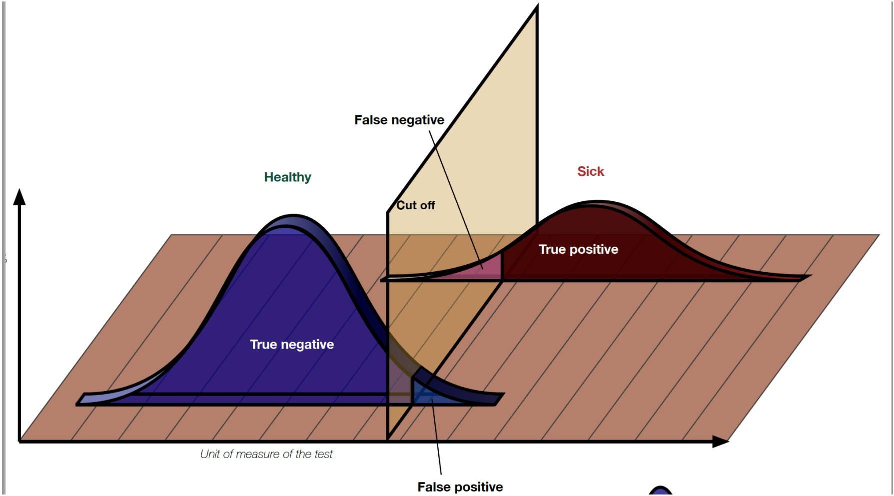
Example (continued): Type I and Type II Errors.

There are two kinds of errors in hypothesis testing: rejecting the null hypothesis H_0 when H_0 is true (type I error) and accepting H_0 when the alternative hypothesis H_1 is true (type II error).

	Accept H_0	Reject H_0
When H_0 is true	Correct Decision	Type I error
When H_1 is true	Type II error	Correct Decision

- **Q: What are the type I and II errors in a Covid-19 antigen rapid test?**
- Typically, there is a trade-off between the type I and type II errors. We cannot minimize the probabilities of type I and type II errors simultaneously.
- The standard approach for testing sets the probability of type I error to be equal to a prespecified level, which is called the significance level of the test. In economics, the significance level is typically chosen to be 10%, 5%, or 1%.

Outline of the Course



Outline of the Course

Statistical Software

- STATA (installed in computer labs at B1 of the HSS building)

Proposed Course Assessment

Grades on this course will be based on the following components.

- Quiz: 20%
- Attendance and Participation: 10%
- Assignment (tutorial questions) and Presentation: 20%
- Final Examination: 50%

Outline of the Course

Main Topics

- Simple Linear Regression Model, Ordinary Least Squares (OLS) Estimator, Least Squares Assumptions
- Properties of the OLS Estimator: Unbiasedness, Efficiency, and Distribution
- Hypothesis Tests and Confidence Intervals for the Linear Regression Model
- Omitted Variable Bias, Linear Regression with Multiple Regressors, Multicollinearity
- Test of Joint Hypothesis, Chi-squared Distribution, F-statistic
- Nonlinear Regression Functions: Polynomial and Logarithmic Models, Interaction Models

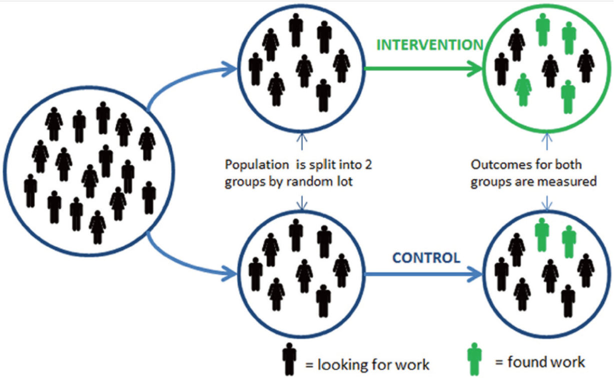
Causality and Econometrics

One effective way of finding the causal effect is to conduct Randomized experiment (randomized controlled trial): Randomly assign the experiment subjects into treatment and control groups; Only the treatment group receives certain treatment, intervention or policy (e.g., only the treatment group receives some job training).

The random assignment makes other things equal on average for the treatment and control groups. Then, we can study the causal effect by simply comparing the difference in outcome between the treatment and control groups, to see whether the policy is effective.

Causality and Econometrics

Example: Randomized experiment for a job training program. The random assignment makes the treatment and control groups very similar, so that comparing the outcome of two groups gives us the "causal effect" of the program/intervention.



Causality and Econometrics

Randomized Experiment and Econometrics

- However, in economics or social science, many randomized experiments are very expensive or even infeasible to conduct (imagine a social planner randomly choose a group of people and assign them certain years of education in order to study the return to education, or randomly assign high-speed railway projects among many cities: These policies are infeasible).
- **Often, we turn to nonexperimental data (in economics, most data are nonexperimental), and still try to reveal the causal effect.**
- Uncovering causal relationship with nonexperimental data is very challenging (e.g., think about the high-speed railway project), but Econometrics bring us as close as possible to the causality-revealing power of a real experiment.

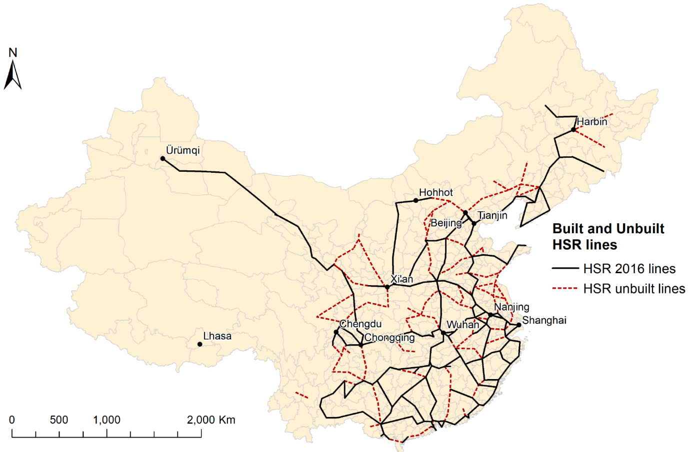
Causality and Econometrics

- More specifically, suppose we would like to study the effects of Chinese high-speed railways (HSR) on regional employment growth.
- Empirical Dataset: Observe 149 planned HSR lines; 83 open by 2016 and 66 not yet open
- To obtain the effect of HSR on employment growth, can we treat the 83 opened lines as a treatment group and the 66 unbuilt lines as a control group (and then compare the employment of the treatment and control groups)?

Causality and Econometrics

HSR Map:

Planned HSR lines



How Does Econometrics Work?

Example: Class Size vs Student Performance.

- Policy question: Suppose that you work for a consultant company, and the superintendent of an elementary school district must decide whether to hire additional teacher, and she wants your advice.
- Parents want smaller classes so that their children can receive more individualized attention. But the superintendent faces a budget constraint.
- **So she asks you: If we cut the class size, what will the causal effect be on student performance in terms of test score?** (e.g., reducing class size by 5 students per class? By 10 students per class?)
- Cutting the class size probably has a positive effect on student performance. But, again, for policy making, we need to know the actual number here.

How Does Econometrics Work?

- We must use real-world data to find out...
- **Data set**: Test score data for all California school districts [Sample size (No. of school districts): $n = 420$];

  Variables:
  - $5th$ grade test scores (Stanford-9 achievement test, combined math and reading), district average,
  - Student-teacher ratio (STR) $=$ No. of students in the district divided by No. of full-time teachers in the district; Therefore, higher STR $=$ larger class size, etc.
- Now the question becomes if we change the student-teacher ratio by a certain amount, what would be the corresponding change in test score?
- How to get the value of such change from the California data set?

How Does Econometrics Work?

Initial Look at the Data:

county	district	testscr	str
Alameda	Sunol Glen Unified	690.8	17.88991
Butte	Manzanita Elementary	661.2	21.52466
Butte	Thermalito Union Elementary	643.6	18.69723
Butte	Golden Feather Union Elementary	647.7	17.35714
Butte	Palermo Union Elementary	640.85	18.67133
Fresno	Burrel Union Elementary	605.55	21.40625
San Joaquin	Holt Union Elementary	606.75	19.5
Kern	Vineland Elementary	609	20.89412
Fresno	Orange Center Elementary	612.5	19.94737
Sacramento	Del Paso Heights Elementary	612.65	20.80556
Merced	Le Grand Union Elementary	615.75	21.23809
Fresno	West Fresno Elementary	616.3	21
Tulare	Allensworth Elementary	616.3	20.6
Tulare	Sunnyside Union Elementary	616.3	20.00822
Tulare	Woodville Elementary	616.45	18.02778
Tulare	Pixley Union Elementary	617.35	20.25196
Kern	Lost Hills Union Elementary	618.05	16.97787
Kern	Buttonwillow Union Elementary	618.3	16.5098
Los Angeles	Lennox Elementary	619.8	22.70402
Kern	Lamont Elementary	620.3	19.91111
Imperial	Westmorland Union Elementary	620.5	18.33333

Initial Look at the Data:

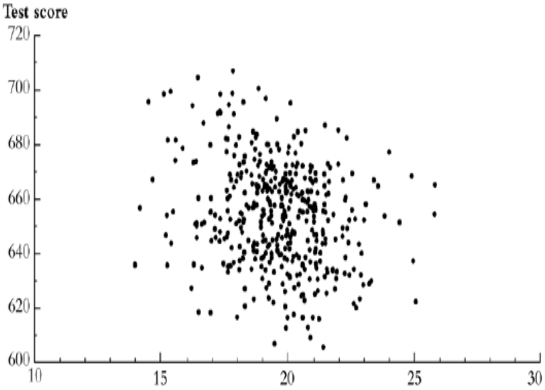
Summary of the Distribution of Student-Teacher Ratios (STR) and Fifth-Grade Test Scores for 420 School Districts in California, US (However, this table does not tell us anything about the relationship between test scores and the STR)

			Percentile						
	Average	Standard Deviation	10%	25%	40%	50% (median)	60%	75%	90%
Student-teacher ratio	19.6	1.9	17.3	18.6	19.3	19.7	20.1	20.9	21.9
Test score	665.2	19.1	630.4	640.0	649.1	654.5	659.4	666.7	679.1

How Does Econometrics Work?

Do districts with smaller classes have higher test scores?

Scatterplot of Test Score vs. Student-Teacher Ratio



What does this figure show? Positive/negative relationship?

How Does Econometrics Work?

First, we can try to write down this as a simple linear relationship:

$$\text{Test Score} = \beta_0 + \beta_1 STR \tag{1}$$

Someone may point out that class size is just one of many factors that may affect the test score (e.g., economic background, school facilities, random reasons related to the performance of the students on the day of the test, ...)

Then, how to model this idea?

$$\text{Test Score} = \beta_0 + \beta_1 STR + \text{other factors} \tag{2}$$

How Does Econometrics Work?

Or, in general, Equation (2) can be written as:

$$Y = \beta_0 + \beta_1 X + u$$

This is called a Simple Linear Regression Model, as it only has one regressor X. β_0 and β_1 are called parameters of the model. u is called error term of the model, as the current dataset does not include factors other than STR. And we are typically more interested in β_1, the slope parameter.

In the example of Class Size vs. Student Performance, we have

$$\begin{aligned} \beta_1 &= \frac{\Delta \text{Test Score}}{\Delta STR} \\ &= \text{change in test score for a unit change in STR} \end{aligned}$$

How Does Econometrics Work?

Suppose that $\beta_1 = -2$, then a reduction in STR (student-teacher ratio) of 2 would yield a change in test score of

$$\Delta \text{Test Score} = \beta_1 \times \Delta STR = (-2) \times (-2) = 4.$$

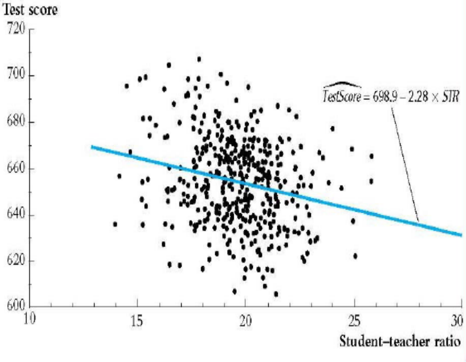
That is to say, smaller class size better student performance.

- However, this result is under the assumption that $\beta_1 = -2$; We do not know the actual value of β_1,
- In practice, it might be positive, negative or even zero,
- So we must estimate it using data (make our best guess of β_1 using data).

How Does Econometrics Work?

California Test Score-Class Size Data (Average test score and STR of 420 school districts, $n = 420$)

- Using appropriate econometric techniques to be learnt in this course, we can find very good estimates of β_0 and β_1.
- For the CA dataset, we have the following estimation result: Test Score = 698.9 - 2.28 $\times$ STR.



How Does Econometrics Work?

Intepretation of the Estimation Result

$$\text{Test Score} = 698.9 - 2.28 \times STR$$

School districts with one unit increase of STR (one more student per teacher) on average have test scores 2.28 points lower. That is,

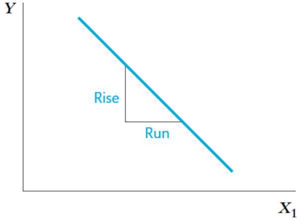
$$\frac{\Delta \text{Test Score}}{\Delta STR} = -2.28,$$

supporting the claim that smaller class size leads to better student performance.

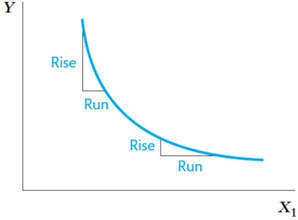
- **Build Econometric Model $\rightarrow$ Estimation with Real Data $\rightarrow$ Intepretation of Estimation Result (and Statistical Testing/Confidence Interval)**

How Does Econometrics Work?

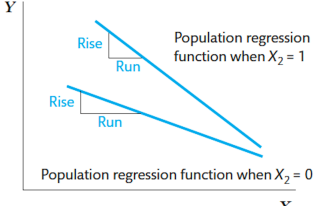
Furthermore, we may also want to incorporate nonlinear relationships: e.g., reducing class size may have a greater effect if class sizes are small; the effect may be larger/smaller for certain group of students.



(a) Constant slope



(b) Slope depends on the value of X_1



The 2021 Nobel Prize in Economics

The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2021

David Card
Joshua D. Angrist
Guido W. Imbens

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11 October 2021

The Royal Swedish Academy of Sciences has decided to award the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2021

with one half to

David Card
University of California, Berkeley, USA

"for his empirical contributions to labour economics"

and the other half jointly to

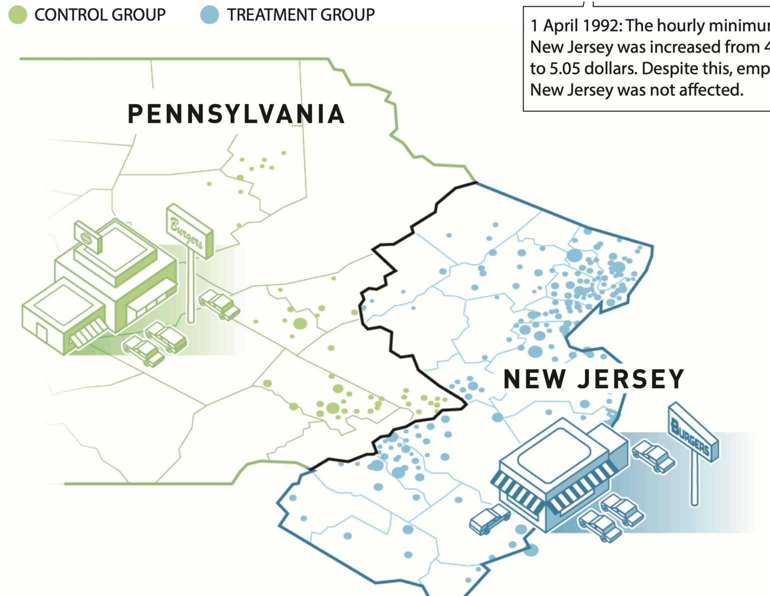
Joshua D. Angrist
Massachusetts Institute of Technology, Cambridge, USA

Application of Econometric Methods to Real-World Data

Press release of the Nobel Prize in Economic Sciences 2021

Using natural experiments, **David Card** has analysed the labour market effects of minimum wages, immigration and education. His studies from the early 1990s challenged conventional wisdom, leading to new analyses and additional insights. The results showed, among other things, that increasing the minimum wage does not necessarily lead to fewer jobs. We now know that the incomes of people who were born in a country can benefit from new immigration, while people who immigrated at an earlier time risk being negatively affected. We have also realised that resources in schools are far more important for students' future labour market success than was previously thought.

Causal Analysis by Using Difference-in-Differences



1 April 1992: The hourly minimum wage in New Jersey was increased from 4.25 dollars to 5.05 dollars. Despite this, employment in New Jersey was not affected.